Wetland Configuration & Event-Scale Flood Response

in the Milwaukee (MMSD) Watersheds

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(1) Event-level (gauge × storm), not annual — capturing the timing/shape mechanisms wetlands actually act on. ▪ **(2) Honest identification:** the wetland *level* effect is confounded (wetland↔urban $r = -0.87$), but the event-varying $W \cdot P$ **interaction is identified** and survives falsification.

1. Background & Baseline

Flooding is the **costliest U.S. natural disaster** ($\sim$\$20B/yr); storms are intensifying, more people live in floodplains, and only $\sim 30\%$ of losses are insured. Wetlands act as **natural sponges**, absorbing floodwater before it reaches homes — yet the U.S. has lost $\sim 40\%$ of its wetlands since 1700. MMSD is restoring wetlands basin-wide on this premise.

Our **anchor paper**, Taylor & Druckenmiller (*AER* 2022), links wetland loss to flood damage through NFIP claims. A direct replication for the Milwaukee basin **failed to identify** a causal effect:

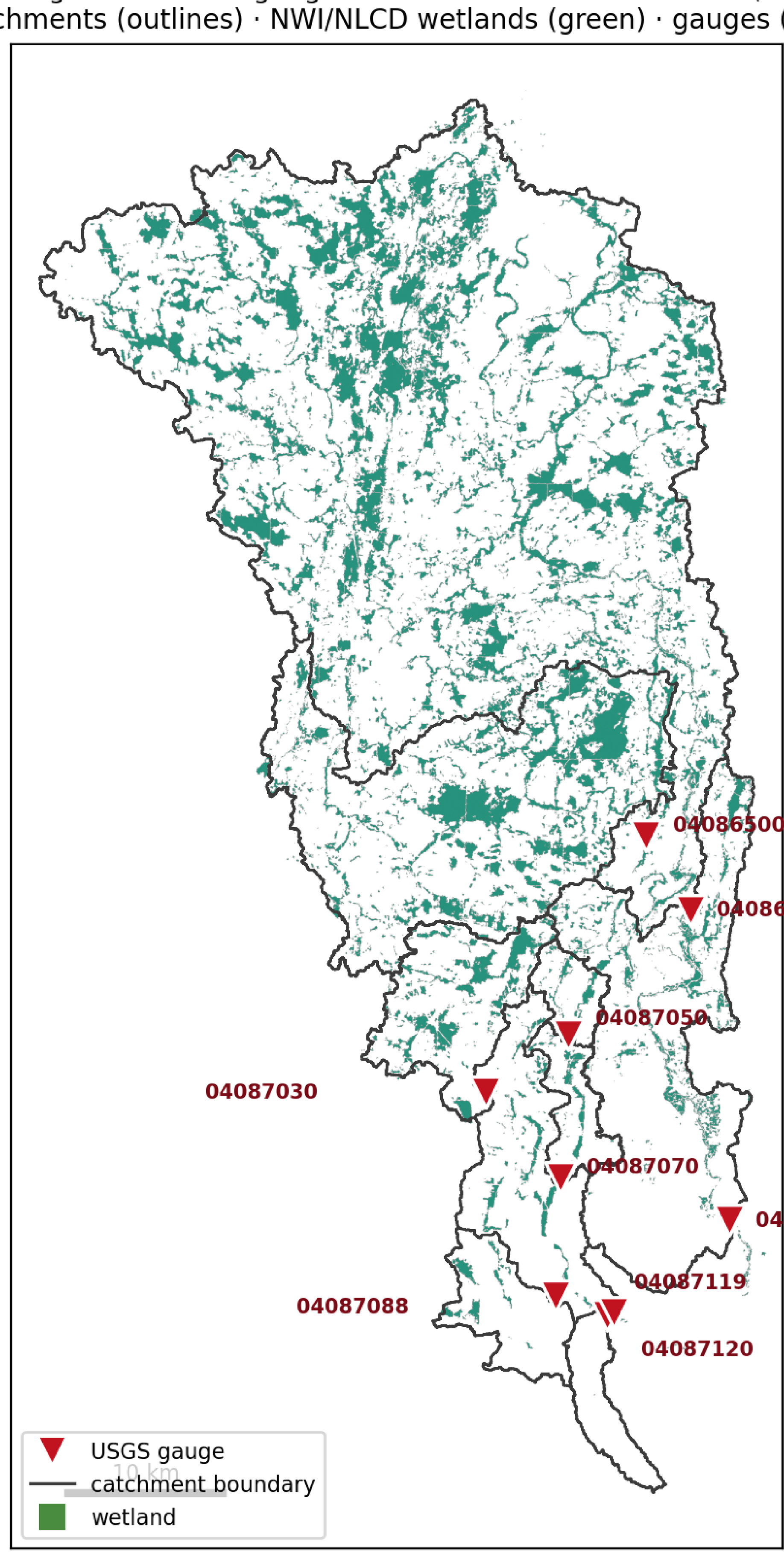
Problem	Detail
No variation	basin wetland area barely changes year to year
Collinearity	wetland loss overlaps urbanization in time/place
ID strategy	gaps in the original identification design

Scope. The full chain is *wetland → flood depth → NFIP damage*; this study does the **hydrologic half** (wetland → event-scale flood response). Depth→damage is future work (§7).

Study area & data

Nine long-record USGS gauges on the Milwaukee & Menomonee rivers, **2001–2023**; **5,907** storm events extracted from 15-minute discharge. The basin grades from a wetland-rich rural north to wetland-poor urban creeks — the gradient that both *powers* and *confounds* the analysis.

Nine long-record USGS gauges — Milwaukee & Menomonee (MMSD) catchments (outlines) · NWI/NLCD wetlands (green) · gauges (red)



At a glance

9 USGS gauges ▪ **5,907** storm events ▪ **2001–2023**

Primary result: the wetland×rainfall interaction on log peak discharge is $\mathbf{-0.20}$ (wild-bootstrap $p = 0.016$) — robust across **four** independent outcomes and all **three** falsification tests, and unchanged under a full rebuild of W.

3. Methodology: a mechanism-based, event-scale model

Hydrology offers two routes: physically-based models (SWAT, HEC-HMS) are explicit but need many non-authoritative, unidentifiable parameters; a purely statistical model is parsimonious but mechanism-blind. We take a **middle path** — regressing observed streamflow responses on rainfall and a compact set of wetland-configuration predictors whose *definitions* are borrowed from the process literature (travel-time connectivity, Volume–Area storage, hydrologic signatures). The model is *phenomenological in estimation, mechanistic in construction*: no invented parameters.

Its defining feature is the **unit of analysis — one gauge × one storm**, not the annual aggregate: wetlands act on the peak, shape, and timing of the hydrograph, which annual means wash out. For event e at gauge j,

$$Y_{j,e} = \beta_0 + \beta_1 P_e + \beta_2 W_j + \beta_3 (W_j \cdot P_e) + \beta_4 (V_e / S_{w,j}) + X_{j,e}\gamma + u_j + \varepsilon_{j,e}$$

Each term is a mechanism: β_1 rainfall forcing; β_2 wetland attenuation; β_3 **peak-shaving** (do wetlands matter more in larger storms?); β_4 saturation (does effectiveness fade once storm volume exceeds capacity?). Correlated predictors are added in a **nested sequence (Models 1–4)**, each mechanism tested against a transparent benchmark.

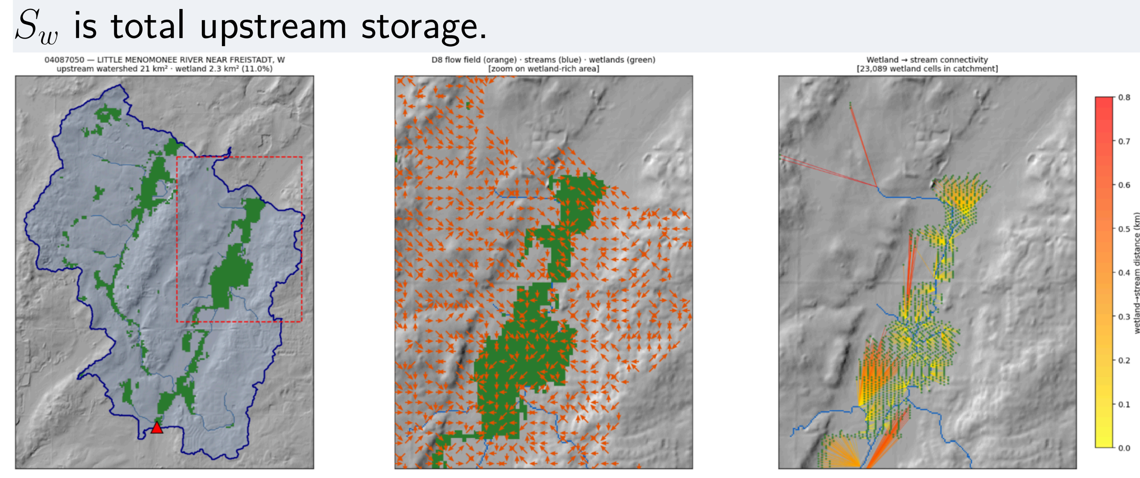
Identification. The wetland *level* effect β_2 is only cross-sectional and confounded; the interaction β_3 is identified *within basins*, across storms of differing size — our inference rests on it.

4-III. Weighted wetland effectiveness W

Not raw area: each wetland contributes **type-weighted storage, discounted by hydrologic travel time** to the gauge, from the advisor form $W_j = \Sigma_i A_i S_i K(T_{ij})$:

$$W_j = \sum_i M_{\text{type},i} \, V_i \, e^{-T_{ij}/\tau}, \qquad S_{w,j} = \sum_i V_i.$$

V_i = storage (NWI area × glaciated Volume–Area law); $M_{\text{type},i}$ = Cowardin-type attenuation weight; T_{ij} = DEM/D8 travel time (kinematic celerity + wetland roughness). The binary connectivity term is **dropped** — near-stream connectivity is already encoded by $e^{-T_{ij}/\tau}$. Thus W is *connectivity-delivered, type-weighted wetland storage*, while S_w is total upstream storage.



Wetlands (green) → D8 flow field → travel-time-weighted connectivity to the gauge.

4-I. Discharge response outcomes Y

From each event hydrograph we build outcomes spanning every mechanism; we model *one at a time.*

Peak (primary): $\log Q_p$ — log of maximum event discharge (tests attenuation).

Relative peak / concentration: *log peak-ratio* (peak vs. local baseline); *log peakedness* (peak excess vs. cumulative quickflow).

Timing / shape: *time-to-peak*; *R–B flashiness*; *hydrograph width*; *recession rate*.

Storage / volume: *runoff coefficient* (quickflow/rainfall); *quickflow depth* (Riemann-sum volume ÷ area).

Extreme severity: Q_{99}*-excess depth* — Riemann-sum volume above the gauge's 99th-percentile flow, area-normalized.

4-II. Rainfall & control variables

P_e = cumulative basin-mean rainfall over the event window (Stage IV / 800 m PRISM radar QPE). **api_30** = 30-day antecedent index, exponentially decayed ($k = 0.9$), summed *strictly before* event start.

P_e and api_30 are built *non-overlapping*: "how big was this storm" vs. "how wet was the basin already."

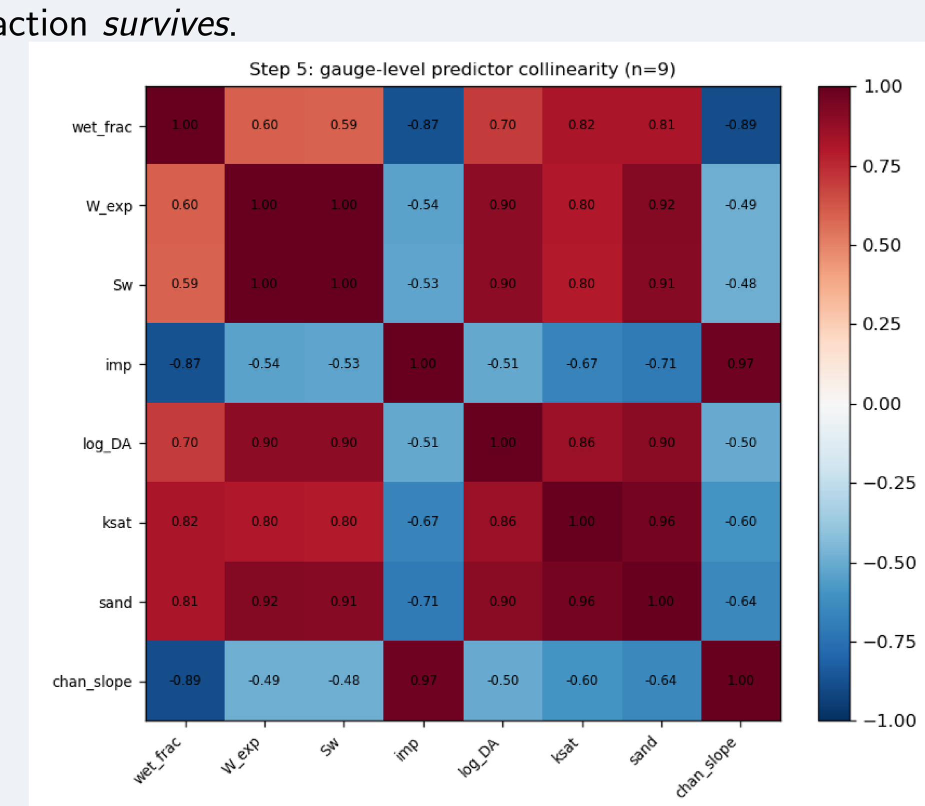
Static controls: log drainage area (size); impervious % (NLCD, urbanization); soil K_{sat} (POLARIS, infiltration); channel slope (3DEP, wave timing); winter flag (snowmelt / cold-season season).

5. Estimation & falsification

OLS with **gauge-clustered SE** (random-effects cross-check), standardized predictors, nested M1–M4. Across nine gauges wetland cover is collinear with urbanization ($r = -0.87$; high VIF for between-gauge terms, but $W \cdot P$ VIF = 1.1). Because nine clusters break asymptotics, every claim is falsified three ways (primary $\log Q_p$):

	Placebo p	Leave-1-out	Wild-boot. p
W (level)	0.09	$[-0.40, -0.04]$	0.16
$W \cdot P$	**0.002**	$[\mathbf{-0.24}, \mathbf{-0.18}]$	**0.016**

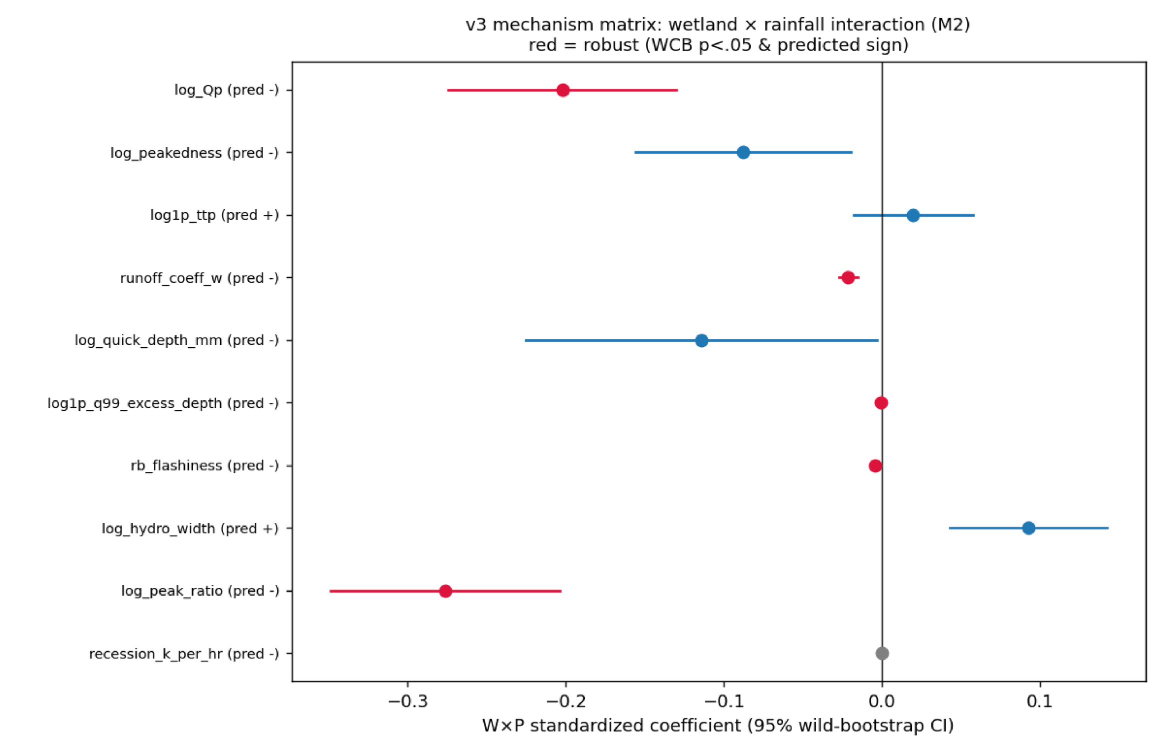
All three agree: the level effect is *not* identified; the peak-shaving interaction *survives*.



Gauge-level collinearity ($n = 9$): wetland vs. urbanization/soils/terrain.

6. Result: wetlands shave flood peaks

More effective upstream wetland storage ⇒ a weaker rainfall-to-peak translation, **disproportionately in larger storms**. On log peak discharge $W \cdot P = \mathbf{-0.20}$ (wild-bootstrap $p = 0.016$), **corroborated** by an independent relative-peak metric and a storage signature.



Outcome	Mechanism	$W \cdot P$ (WCB p)
$\log Q_p$	peak	$\mathbf{-0.20}$ (.016)
log peak-ratio	rel. peak	$\mathbf{-0.28}$ (.027)
runoff coef.	storage	$\mathbf{-0.02}$ (.029)
flashiness	shape	$\mathbf{-0.004}$ (.043)
peakedness / width	shape/time	pred. sign, n.s.

9 of 10 outcomes carry the predicted sign; peak-shaving is corroborated by a second peak metric and a storage signature, and is **invariant** to a full, literature-grounded rebuild of W (v1→v3) and to correcting the antecedent control.

7. Conclusion, limitations & future work

Wetlands **reshape the storm hydrograph's peak**, more so in larger storms — a mechanistically grounded, falsification-surviving *association*. The wetland level effect is **not causally identified**: nine nested, urbanization-confounded gauges limit us to within-basin variation; absolute travel times are uncalibrated (so W is a relative index) and S_w is order-of-magnitude.

Future work. (i) a regional sample of *independent* catchments spanning a wetland gradient decoupled from urbanization, to identify the level effect; (ii) native 4 km/1 km radar over the smallest basins; (iii) close the chain by linking event flood response to **flood depth and NFIP damages** (the economic half, mirroring §1).

References. Acreman & Holden (2013); Ameli & Creed (2019); Lane et al. (2018); Wu & Lane (2016); Kadlec (1990); Taylor & Druckenmiller (2022).

Bottom line: wetlands don't stop the water — they **slow and reshape** it, shaving the flood peak most when storms are largest.